PLSC 504: Fall 2024 Panel Data: Causal Inference + GLMs

October 12, 2026

Things We Can Always Do

Whether or not we have panel data, we can use:

- Regression!
- Matching...
- Instrumental variables...
- Regression discontinuity (RDD)...

Panel Data Approaches: Differences-In-Differences

Simple "DiD":

- Leverages two-group, two-period data ($T = 2$):

	Pre-Treatment ($T = 0$)	Post-Treatment ($T = 1$)
Treated ($W = 1$)	A	B
Untreated ($W = 0$)	C	D

- Process (simple version):
  - Calculate pre- vs. post-treatment difference for the treated group ($B - A$)
  - Calculate pre- vs. post-treatment difference for the untreated group ($D - C$)
  - Calculate the differences between the differences [$\widehat{\text{DiD}} = (B - A) - (D - C)$]

Differences-In-Differences (continued)

Simple DiD is the same as fitting the regression:

$$Y_{it} = \beta_0 + \beta_1 W_{it} + \beta_2 T_{it} + \beta_3 W_{it} T_{it} + u_{it}$$

- Validity depends on (a) all the usual assumptions required by OLS, plus (b) the *parallel trends assumption* – that there are no (non-treatment-related) time-varying differences between the two groups as we go from $T = 0$ to $T = 1$.

- Resources:
  - Our old friend Wikipedia
  - Pischke's slides on DiD
  - R: package `did`
  - Stata: `ieddtab` in the `ietoolkit`

Panel Data Approaches: Synthetic Controls

The "synthetic control method" (SCM):

- Addresses situations in which we have a *single* treated case (or small number of them)...
- Requires at least one (and ideally more) repeated measurements over time on the outcome of interest; works best with reasonably long T both before and after the treatment
- Also requires multiple (but not *too* many) non-treated cases
- Assumptions:
  - Possible control units are similar
  - Lack of spillover between treated and potential control units
  - Lack of exogenous shocks to potential control units

Synthetic Controls (continued)

SCM details:

- Intuition:
  - Create a counterfactual "control" unit that is as similar to the (pre-treatment) treated case as possible
  - Do so by weighting the observed predictors across "control" cases to minimize the difference (in a MSE sense)
  - Compare the pre-treatment trends in the synthetic control and treated cases
  - The weights are then used to create a post-treatment trend for the synthetic control
  - Inference is via placebo methods (varying the timing of the intervention)
- Advantages:
  - Works with (very) small N
  - Doesn't require parallel trends (a la DiD)
  - Abadie et al. claim that SCM controls for both observed and unobserved time-varying confounders
- A few references:
  - A nontechnical introduction in the *BMJ*
  - Method of the Month Blog
  - The Development Impact blog post on SCM

Software Matters

In general:

- R
  - Packages for matching are listed above (`Matching`, `MatchIt`, etc.)
  - Similarly for RDD (`rddtools`, `rdd`, etc.) and DiD (`did`)
  - IV regression: `ivreg` (in `AER`), `tsls` (in `sem`), others
  - Synthetic controls are in `Synth` and `MicroSynth`
  - See generally the CRAN Task View on *Causal Inference*.
- Stata also has a large suite of routines for attempting causal inference with observational data...
- And there's a pretty good NumPy/SciPy-dependent package for Python, called (creatively) Causalinference

Causal Inference: One-Way (FE) Models

Imai and Kim (2019):

- The punch line first: "(t)he ability of unit fixed effects regression models to adjust for unobserved time-invariant confounders comes at the expense of dynamic causal relationships between treatment and outcome variables."
- Also dependent on functional form assumptions (specifically, linearity)

Intuition: For the model:

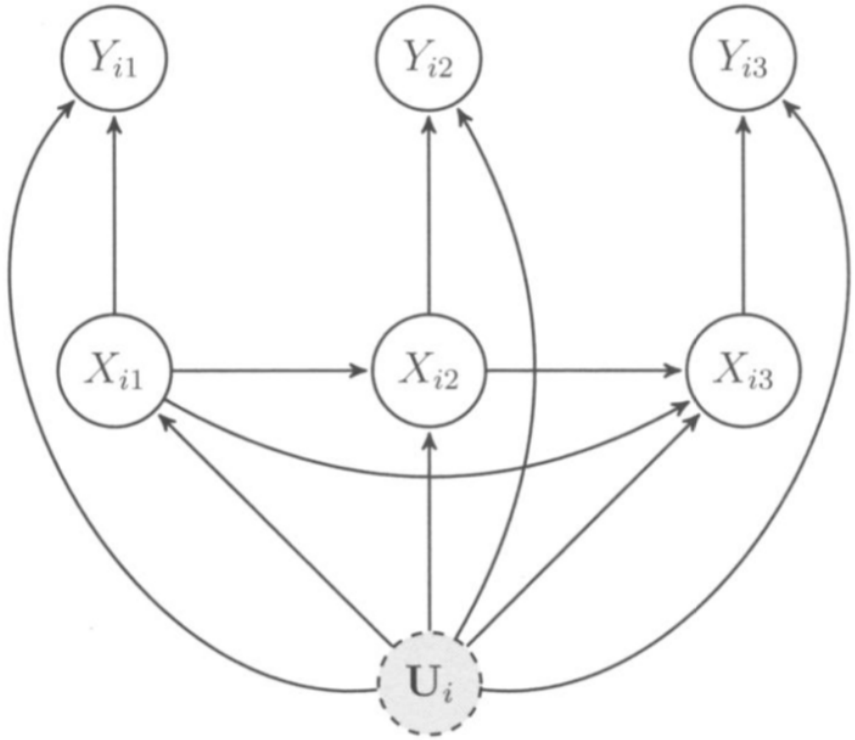
$$Y_{it} = \mathbf{X}_{it}\boldsymbol{\beta} + \alpha_i + u_{it}$$

where (for simplicity) X is a binary treatment for which we want to know a causal effect on Y:

- Identification is via $\text{Cov}[(\mathbf{X}_{it}, \alpha_i), u_{it}] = 0$
- In this framework, $\boldsymbol{\beta} = \tau$, the typical causal estimand (that is, the expected difference between $Y_{it}(0)$ and $Y_{it}(1)$)

A more flexible approach is to think of a FE model as a DAG...

Fixed-Effects DAG



Source: Imai and Kim (2019).

Key FE Takeaways

Summarizing Imai and Kim (2019):

- Three key identifying assumptions for FE models:
  - No unobserved time-varying confounders
  - Past treatments / values of $\mathbf{X}$ do not affect current values of Y[1]
  - Past outcomes Y do not affect current values of $\mathbf{X}$.
- Alternatively, one can select on observables (a la Blackwell and Glynn 2018) and model dynamics (albeit at the cost of failing to control for unobserved time-constant confounders).

> *"...researchers must choose either to adjust for unobserved time-invariant confounders through unit fixed effects models or to model dynamic causal relationships between treatment and outcome under a selection-on-observables approach. No existing method can achieve both objectives without additional assumptions" (Imai and Kim 2019, 484).*

[1] Can be relaxed via IV, but that requires independence of past and present values of Y.

Imai and Kim redux (2020):

- In the simple $T = 2$ case, DiD is equivalent to a two-way FE model:

$$Y_{it} = \mathbf{X}_{it}\boldsymbol{\beta} + \alpha_i + \eta_t + u_{it}$$

- Imai & Kim: The same is not true for $T > 2$...
- More important: two-way FEs' ability to control for unmeasured confounders depends on the (linearity of the) functional form...
- Upshot: two-way FEs aren't a (nonparametric) cure-all...
- Related: When we control for both α_i and η_t, what – exactly – is the counterfactual?

Back To The WDI

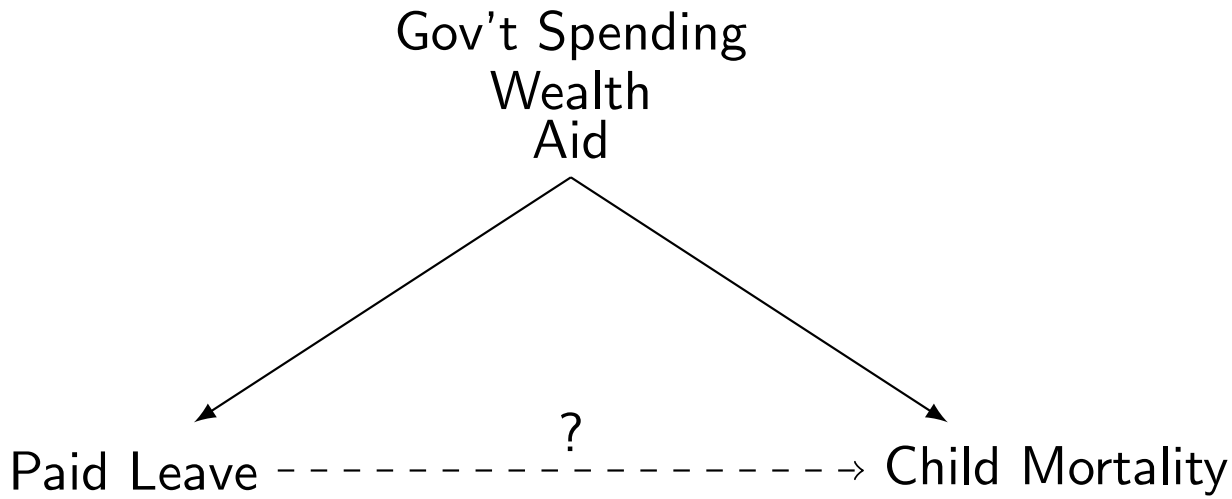
```
> describe(WDI,fast=TRUE,ranges=FALSE,check=TRUE)
```

	vars	n	mean	sd	skew	kurtosis	se
ISO3	1	14298	NaN	NA	NA	NA	NA
Year	2	14298	NaN	NA	NA	NA	NA
Region	3	14124	NaN	NA	NA	NA	NA
country	4	14190	NaN	NA	NA	NA	NA
iso3c	5	14190	NaN	NA	NA	NA	NA
RuralPopulation	6	13975	47.79	25.80	-0.09	-0.99	0.22
UrbanPopulation	7	13975	52.21	25.80	0.09	-0.99	0.22
BirthRatePer1K	8	13945	27.51	13.19	0.25	-1.23	0.11
FertilityRate	9	13943	3.80	1.99	0.45	-1.17	0.02
PrimarySchoolAge	10	11539	6.13	0.61	-0.03	0.13	0.01
LifeExpectancy	11	13941	65.31	11.25	-0.77	0.19	0.10
AgeDepRatioOld	12	13945	10.84	7.27	1.80	5.05	0.06
ChildMortality	13	11821	72.07	78.05	1.66	2.97	0.72
GDP	14	11307	243922334825.73	1153249695108.14	11.54	161.13	10845501840.29
GDPPerCapita	15	11307	12443.80	19449.79	3.17	15.21	182.91
GDPPerCapGrowth	16	11184	1.91	6.54	1.68	40.87	0.06
TotalTrade	17	8802	78.15	53.67	3.01	17.96	0.57
FDIIn	18	9198	5.46	44.41	15.43	561.70	0.46
NetAidReceived	19	9218	528849286.43	1113709450.10	11.21	269.53	11599881.51
MobileCellSubscriptions	20	10587	39.15	53.62	1.26	1.61	0.52
NaturalResourceRents	21	9211	6.84	11.06	2.60	8.04	0.12
GovtExpenditures	22	8470	16.39	8.26	3.65	32.91	0.09
PublicHealthExpend	23	4547	3.31	2.40	1.37	3.43	0.04
WomenInLegislature	24	5259	18.62	12.10	0.66	-0.04	0.17
PostColdWar	25	14190	0.55	0.50	-0.18	-1.97	0.00
lnGDPPerCap	26	11307	8.40	1.52	0.08	-0.89	0.01
lnNetAidReceived	27	9051	18.85	1.97	-1.07	2.08	0.02
YearNumeric	28	14190	1992.50	19.05	0.00	-1.20	0.16
WomenBusLawIndex	29	10260	59.97	18.75	0.01	-0.59	0.19
PaidParentalLeave	30	10260	0.11	0.31	2.50	4.24	0.00

A New Question

Do paid parental leave policies decrease child mortality?

- Y = `ChildMortality` (N of deaths of children under 5 per 1000 live births) **(logged)**
- T = `PaidParentalLeave` (1 if provided, 0 if not)
- **X**s:
  - `GDPPerCapita` (Wealth; in constant $US) (logged)
  - `NetAidReceived` (Net official development aid received; in constant $US) (logged)
  - `GovtExpenditures` (Government Expenditures, as a percent of GDP)



Preliminary Regressions

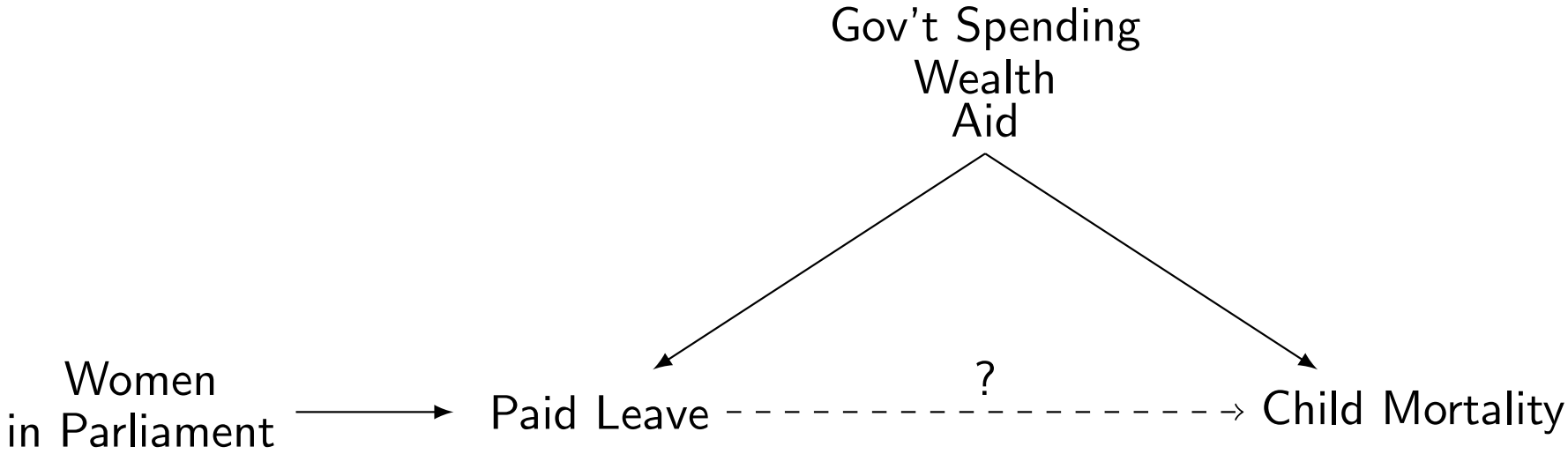
Table: Models of log(Child Mortality)

	Bivariate OLS	OLS	One-Way FE	Two-Way FE	FE w.Lagged Y
Paid Parental Leave	−1.850	−0.930***	−0.132***	−0.159***	−0.229***
	(0.033)	(0.038)	(0.043)	(0.024)	(0.028)
ln(GDP Per Capita)		−0.603***	−1.070***	−0.265***	−0.572***
		(0.009)	(0.016)	(0.012)	(0.012)
ln(Aid Received)		−0.004	−0.018***	0.005***	0.002
		(0.003)	(0.002)	(0.001)	(0.002)
Government Expenditures		0.0002	0.003***	0.0004	0.0003
		(0.001)	(0.001)	(0.001)	(0.001)
Lagged Child Mortality					0.008***
					(0.0001)
Constant	3.730*	8.680***			
	(0.011)	(0.112)			
Observations	9,814	5,396	5,396	5,396	5,392
R^2	0.238	0.554	0.472	0.122	0.787
Adjusted R^2	0.238	0.554	0.457	0.087	0.780

*p<0.1; **p<0.05; ***p<0.01

Instrumental Variables

Conceptually:



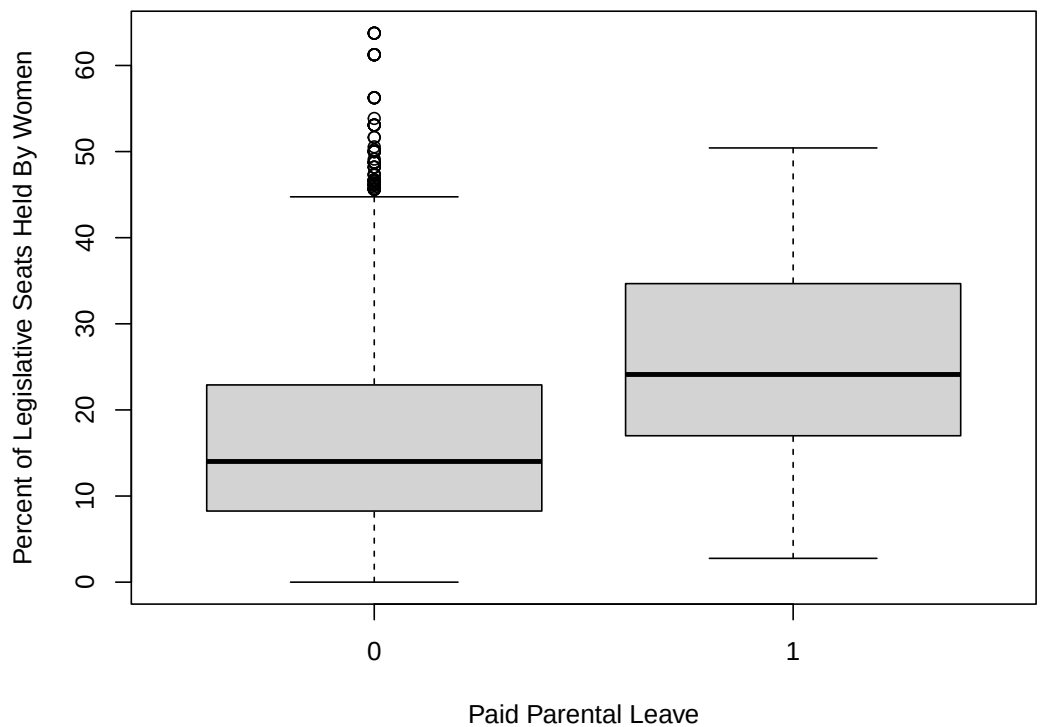
Instrumental Variables (continued)

Assessing Cov($W, \mathbf{Z}$):

```
> with(WDI,t.test(WomenInLegislature~PaidParentalLeave))

 Welch Two Sample t-test

data:  WomenInLegislature by PaidParentalLeave
t = -22, df = 1467, p-value <0.0000000000000002
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
95 percent confidence interval:
 -9.78 -8.21
sample estimates:
mean in group 0 mean in group 1
           16.5            25.5
```



Instrumental Variables: Syntax

E.g., one-way fixed effects with IV:

```
FE.IV<-plm(lnCM~PaidParentalLeave+log(GDPPerCapita)+
             log(NetAidReceived)+GovtExpenditures |
             . - PaidParentalLeave+WomenInLegislature,
             data=WDI,effect="individual",model="within")
```

Instrumental Variable Results

Table: IV Models of log(Child Mortality)

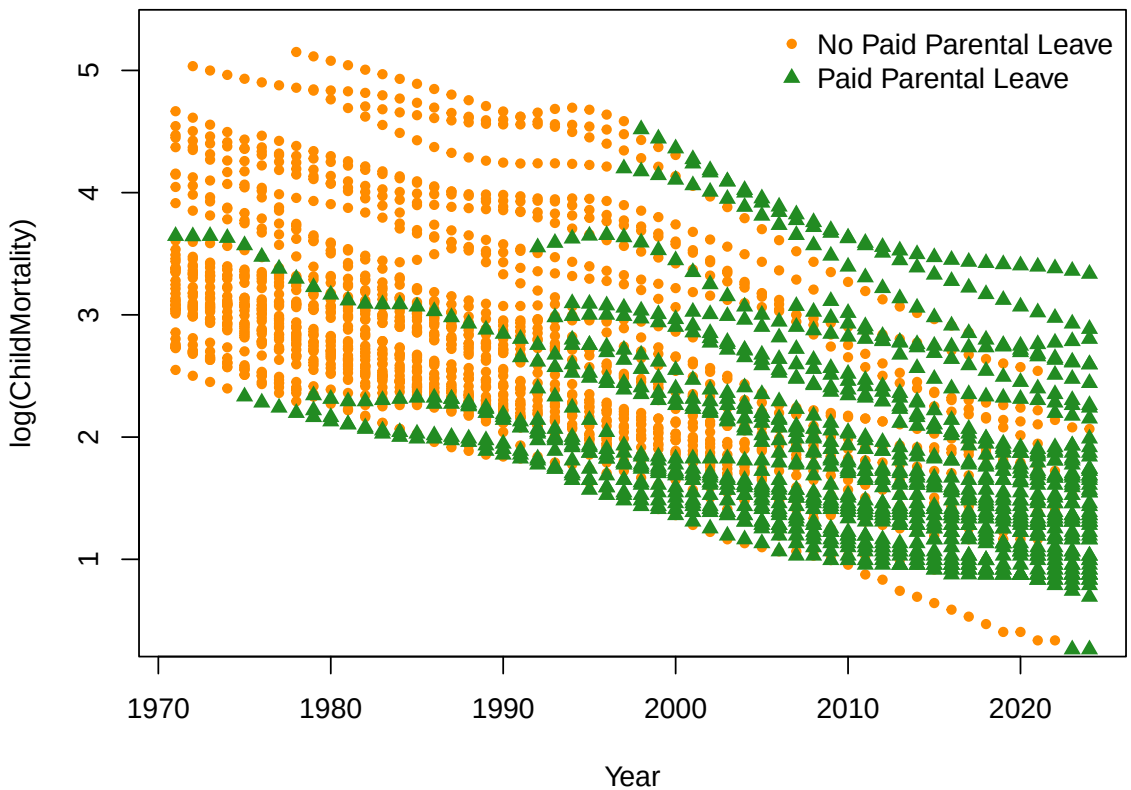
	OLS	One-Way FE	FE w/IV	RE w/IV
Paid Parental Leave	−0.930	−0.132***	89.200	−5.530***
	(0.038)	(0.043)	(194.000)	(1.540)
ln(GDP Per Capita)	−0.603	−1.070***	−13.900	−0.459***
	(0.009)	(0.016)	(27.900)	(0.067)
ln(Aid Received)	−0.004	−0.018***	0.072	0.021*
	(0.003)	(0.002)	(0.204)	(0.012)
Government Expenditures	0.0002	0.003***	0.049	−0.001
	(0.001)	(0.001)	(0.129)	(0.003)
Constant	8.680*			7.210***
	(0.112)			(0.565)
Observations	5,396	5,396	2,919	2,919
R^2	0.554	0.472	0.00003	0.243
Adjusted R^2	0.554	0.457	-0.053	0.242

*p<0.1; **p<0.05; ***p<0.01

Intuition: Compare the child mortality "trajectories" of countries before and after they implement paid parental leave policies.

The model is:

$$\begin{aligned} \text{Child Mortality}_{it} &= \beta_0 + \beta_1(\text{Paid Parental Leave}_{it}) + \beta_2(\text{Time}_t) + \\ &= \beta_3(\text{Paid Parental Leave}_{it} \times \text{Time}_t) + (\text{confounders}) + u_{it} \end{aligned}$$



RDD Regressions

Table 3: RDD Models of log(Child Mortality)

	OLS #1	OLS #2	One-Way FE #1	One-Way FE #2	Two-Way FE #1	Two-Way FE #2
(Intercept)	4.4686***	10.2516***				
	(0.1335)	(0.1865)				
Paid Parental Leave	−0.4374	−0.0970	0.1108	0.0554	0.1561	0.0665
	(0.3039)	(0.1864)	(0.0857)	(0.0768)	(0.1147)	(0.1062)
Time (1950=0)	−0.0321***	−0.0246***	−0.0520***	−0.0425***		
	(0.0028)	(0.0016)	(0.0007)	(0.0011)		
Paid Parental Leave x Time	0.0086	−0.0005	−0.0023	−0.0011	−0.0034+	−0.0015
	(0.0054)	(0.0033)	(0.0014)	(0.0013)	(0.0020)	(0.0019)
ln(GDP Per Capita)		−0.6678***		−0.2243***		−0.2174***
		(0.0172)		(0.0230)		(0.0251)
ln(Aid Received)		−0.0017		0.0068***		0.0061***
		(0.0035)		(0.0014)		(0.0015)
Government Expenditures		−0.0269***		0.0085***		0.0087***
		(0.0032)		(0.0016)		(0.0017)
Num.Obs.	825	825	825	825	825	825
R2	0.192	0.740	0.919	0.936	0.009	0.205
R2 Adj.	0.189	0.738	0.915	0.933	−0.105	0.110

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Challenges:

- Multiple periods (years) per unit (country), both before and after "treatment"
- "Staggered" treatment timing (adoption of *Paid Parental Leave*)

One approach:

Callaway, Brantley, and Pedro H.C. Sant'Anna. 2021. "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics* 225:200-230.

Details:

- Deals with the issues related above
- Flexibly fit / interpreted using the `did` package

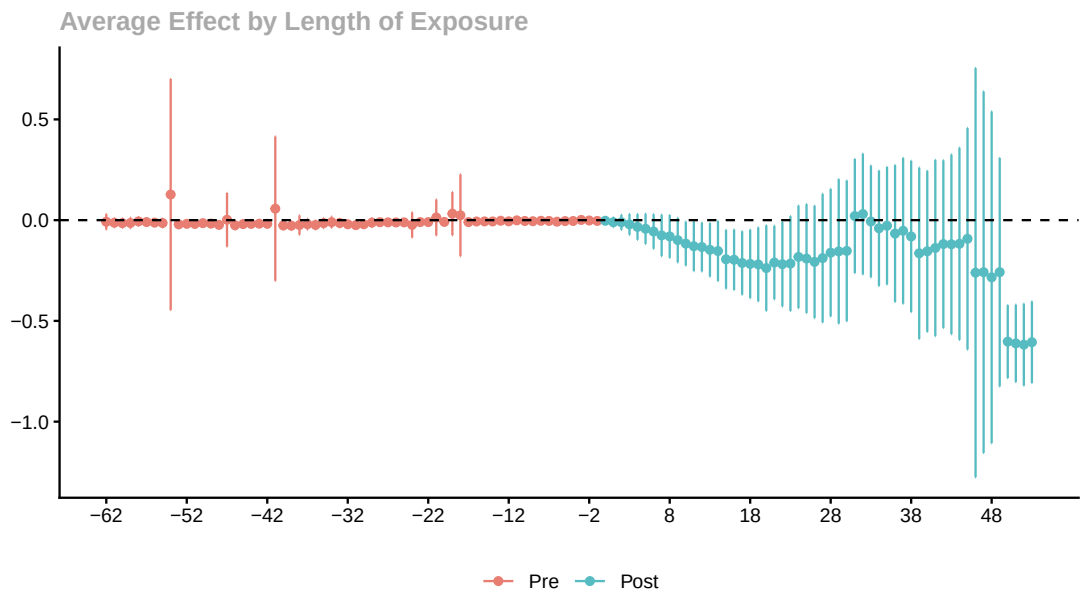
Differences-in-Differences via did

Simple bivariate model (no controls):

```
> DiD.fit1<-att_gt(yname = "lnCM",gname = "YearPPL",idname = "ID",
+                  tname = "YearNumeric",allow_unbalanced_panel = TRUE,
+                  xformla = ~1,data = WDI,est_method = "reg")

> # Event study object:
>
> DiD.ev1 <- aggte(DiD.fit1,type="dynamic",na.rm=TRUE)
```

Plot the event study results:



ATTs by "Group"

```
> DiD.grp1<-aggte(DiD.fit1,type="group",na.rm=TRUE)
> summary(DiD.grp1)

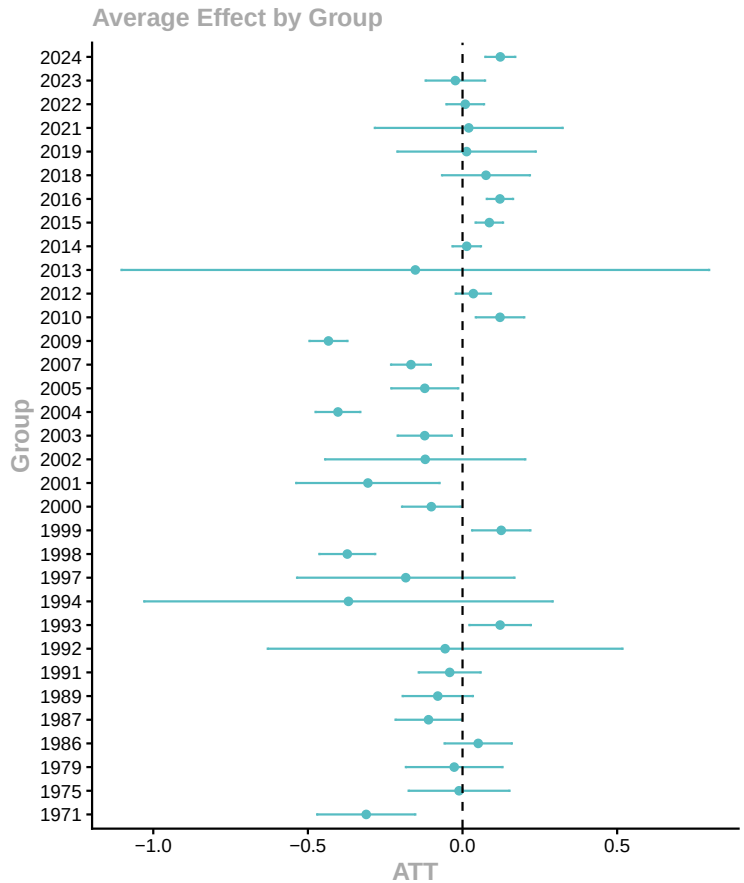
Call:
aggte(MP = DiD.fit1, type = "group", na.rm = TRUE)

Reference: Callaway, Brantly and Pedro H.C. Sant'Anna.
"Difference-in-Differences with Multiple Time Periods."
Journal of Econometrics, 225(2) pp. 200-230, 2021.

Overall summary of ATT's based on group/cohort aggregation:
     ATT    Std. Error     [ 95%  Conf. Int.]
 -0.0791        0.0285     -0.135     -0.0232 *


Group Effects:
 Group Estimate Std. Error [95% Simult.  Conf. Band]
  1971  -0.3114     0.0376       -0.4144     -0.2084 *
  1975  -0.0112     0.0406       -0.1224      0.1000
  1979  -0.0269     0.0426       -0.1438      0.0900
  1986   0.0507     0.0262       -0.0212      0.1226
  1987  -0.1100     0.0244       -0.1769     -0.0430 *
  1989  -0.0801     0.0244       -0.1468     -0.0133 *
  1991  -0.0415     0.0227       -0.1037      0.0208
  1992  -0.0562     0.1370       -0.4317      0.3192
  .
  .
  .
  2021   0.0201     0.0759       -0.1879      0.2281
  2022   0.0088     0.0160       -0.0352      0.0528
  2023  -0.0232     0.0241       -0.0893      0.0429
  2024   0.1222     0.0119        0.0894      0.1549 *
---
Signif. codes: '*' confidence band does not cover 0

Control Group:  Never Treated,  Anticipation Periods:  0
Estimation Method:  Outcome Regression
```

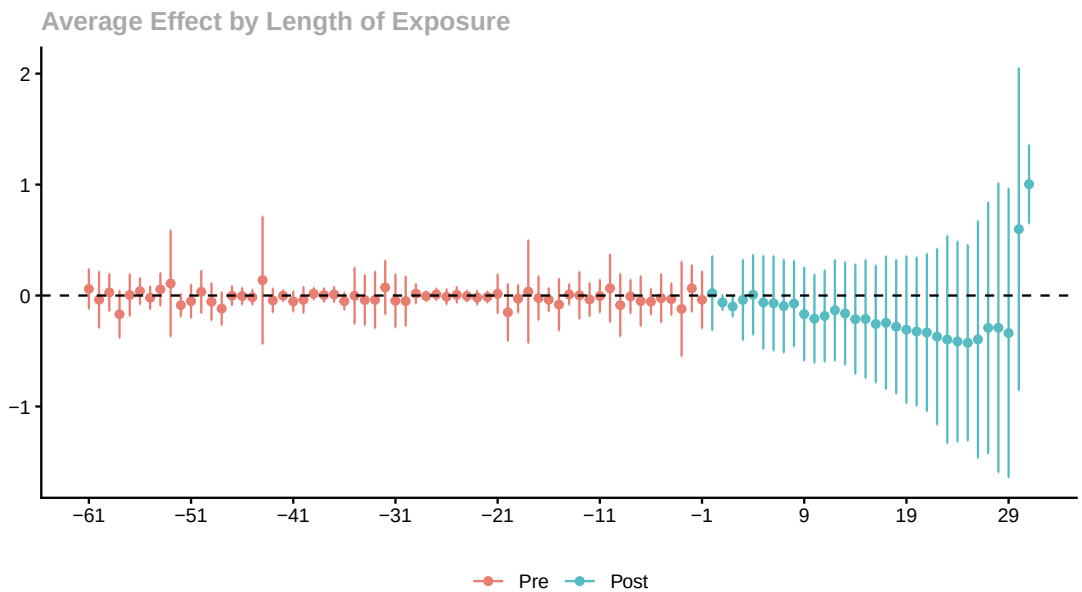


Differences-in-Differences with Controls

Adding control variables:

```
> DiD.fit2<-att_gt(yname = "lnCM",gname = "YearPPL",idname = "ID",
+                  tname = "YearNumeric",allow_unbalanced_panel = TRUE,
+                  xformla = ~lnGDPPerCap+lnNetAidReceived+GovtExpenditures,
+                  data = WDI, est_method = "reg")
>
> # Event study object:
>
> DiD.ev2 <- aggte(DiD.fit2,type="dynamic",na.rm=TRUE)
```

Plot the event study results:



ATTs by "Group" (with controls)

```
> DiD.grp2<-aggte(DiD.fit2,type="group",na.rm=TRUE)
> summary(DiD.grp2)

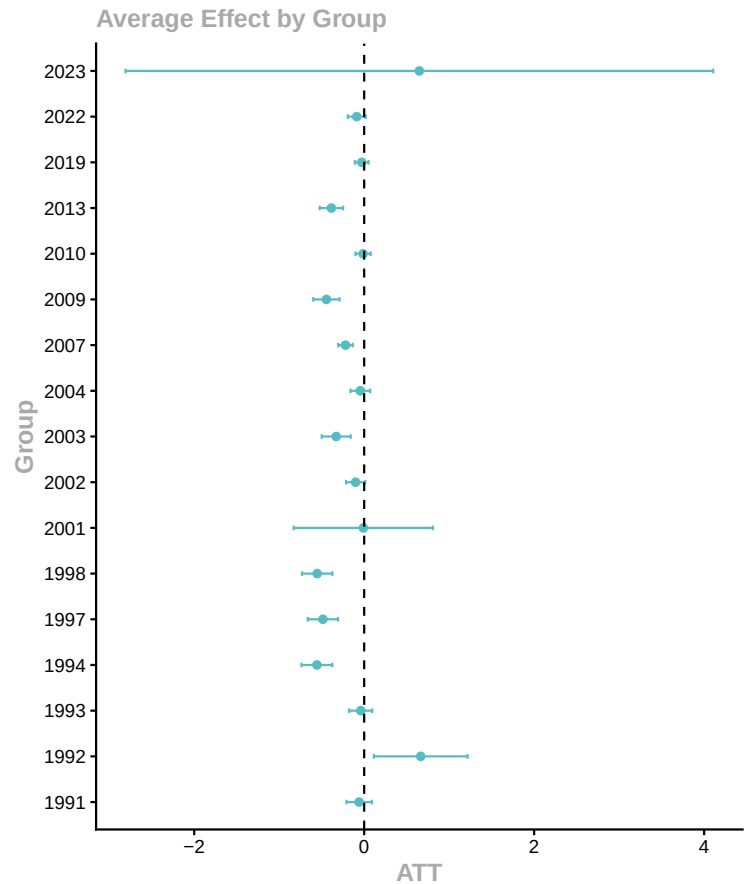
Call:
aggte(MP = DiD.fit2, type = "group", na.rm = TRUE)

Reference: Callaway, Brantly and Pedro H.C. Sant'Anna.
"Difference-in-Differences with Multiple Time Periods."
Journal of Econometrics, 225(2), pp. 200-230, 2021.

Overall summary of ATT's based on group/cohort aggregation:
    ATT    Std. Error     [ 95%  Conf. Int.]
 -0.159        0.0815     -0.319      0.0006


Group Effects:
 Group Estimate Std. Error [95% Simult.  Conf. Band]
  1991  -0.0786     0.0657        -0.258      0.1007
  1992   0.6752     0.1626         0.231      1.1190 *
  1993   0.0200     0.0604        -0.145      0.1848
  1994  -0.5053     0.0710        -0.699     -0.3115 *
  1997  -0.5170     0.0723        -0.714     -0.3197 *
  1998  -0.5866     0.0648        -0.764     -0.4096 *
  2001  -0.0500     0.3584        -1.028      0.9284
  2002  -0.0779     0.0448        -0.200      0.0443
  2003  -0.2668     0.0902        -0.513     -0.0206 *
  2004  -0.0751     0.0277        -0.151      0.0005
  2007  -0.2449     0.0342        -0.338     -0.1515 *
  2009  -0.4643     0.0574        -0.621     -0.3077 *
  2010  -0.0575     0.0386        -0.163      0.0478
  2013  -0.3860     0.0544        -0.534     -0.2374 *
  2019  -0.0105     0.0363        -0.110      0.0887
  2022  -0.0933     0.0433        -0.211      0.0247
---
Signif. codes: '*' confidence band does not cover 0

Control Group:  Never Treated,  Anticipation Periods:  0
Estimation Method:  Outcome Regression
```



Resources

Good references:

- Freedman (2012)*
- Shalizi (someday)*
- Morgan and Winship (2014)
- Pearl et al. (2016)
- Peters et al. (2017)

Other useful things:

- The CRAN task view on causal inference
- The What If Book
- Some useful notes

Courses / syllabi (a sampling):

- Blackwell (2023)
- Eggers (2019)
- Frey (2024)
- Imai (2023)
- Munger (2023)
- Ornstein (2023)
- Xu (2018, 2023).
- Yamamoto (2021)

* I really like this one.

“Generalized” Models

Unit Effects, Fixed

One-way unit effects:

$$Y_{it} = f(\mathbf{X}_{it}\beta + \alpha_i + u_{it})$$

So, think about logit first:

$$\Pr(Y_{it} = 1) = \frac{\exp(\mathbf{X}_{it}\beta + \alpha_i)}{1 + \exp(\mathbf{X}_{it}\beta + \alpha_i)} \equiv \Lambda(\mathbf{X}_{it}\beta + \alpha_i)$$

Incidental Parameters:

- Nonlinearity $\rightarrow$ inconsistency in both $\hat{\alpha}$s *and* $\hat{\boldsymbol{\beta}}$.
- Anderson's *unconditional* estimator:

$$L^U = \prod_{i=1}^{N}\prod_{t=1}^{T} \Lambda(\mathbf{X}_{it} + \alpha_i)^{Y_{it}}[1 - \Lambda(\mathbf{X}_{it} + \alpha_i)]^{1-Y_{it}}$$

- Chamberlain's *conditional* estimator:

$$L^C = \prod_{i=1}^{N} \Pr\left(Y_{i1} = y_{i1}, Y_{i2} = y_{i2}, \ldots Y_{iT} = y_{iT} \mid \sum_{t=1}^{T} Y_{it}\right)$$

Fixed-Effects (continued)

Intuition: Suppose we have $T = 2$. That means that:

- $\Pr(Y_{i1} = 0 \text{ and } Y_{i2} = 0 \mid \sum_T Y_{it} = 0) = 1.0$
- $\Pr(Y_{i1} = 1 \text{ and } Y_{i2} = 1 \mid \sum_T Y_{it} = 2) = 1.0$

and:

$$\Pr\left(Y_{i1} = 0 \text{ and } Y_{i2} = 1 \mid \sum_T Y_{it} = 1\right) = \frac{\Pr(0,1)}{\Pr(0,1) + \Pr(1,0)}$$

with a similar statement for $\Pr(Y_{i1} = 1 \text{ and } Y_{i2} = 0 \mid \sum_T Y_{it} = 1)$.

The Point:

$\sum_{t=1}^{T} Y_{it}$ **is a sufficient statistic for** α_i, so conditioning on it $\equiv$ "fixed effects."

Notes On Fixed-Effects

Things to bear in mind:

- Fixed effects = no estimates for $\boldsymbol{\beta}_b$
- Interpretation: per logit, but $\mid \hat{\alpha}_i$.
- Everything above is for **logit**...
  - For FE probit, there is no conditional model
  - Unconditional / "brute force" FE probit is biased (see here and here)
- BTSCS in international relations: Green et al. (2001) vs. Beck & Katz (2001) ("Dirty Pool" debate)

Random Effects

Model is:

$$\begin{aligned} Y_{it}^{*} &= \mathbf{X}_{it}\beta + u_{it} \\ Y_{it} &= 0 \text{ if } Y_{it}^{*} \leq 0 \text{ ;} \\ &= 1 \text{ if } Y_{it}^{*} > 0 \end{aligned}$$

with:

$$u_{it} = \alpha_i + \eta_{it}$$

with $\eta_{it} \sim$ i.i.d. N(0,1), and $\alpha_i \sim N(0, \sigma_\alpha^2)$. This implies:

$$\text{Var}(u_{it}) = 1 + \sigma_\alpha^2$$

and so:

$$\text{Corr}(u_{it}, u_{is},\ t \neq s) \equiv \rho = \frac{\sigma_\alpha^2}{1 + \sigma_\alpha^2}$$

which means that we can write $\sigma_\alpha^2 = \left(\frac{\rho}{1-\rho}\right)$.

Random Effects Variants

Probit:

$$\begin{aligned} L_i &= \text{Prob}(Y_{i1} = y_{i1},\ Y_{i2} = y_{i2},\ ...Y_{iT} = y_{iT}) \\ &= \int_{-\infty}^{x_{i1}\beta} \int_{-\infty}^{x_{i2}\beta} ... \int_{-\infty}^{x_{iT}\beta} \phi(u_{i1}, u_{i2}...u_{iT}) du_{iT}...du_{i2}\ du_{i1} \end{aligned}$$

Logit:

$$\begin{aligned} L_i &= \text{Prob}(Y_{i1} = y_{i1},\ Y_{i2} = y_{i2},\ ...Y_{iT} = y_{iT}) \\ &= \int_{-\infty}^{x_{i1}\beta} \int_{-\infty}^{x_{i2}\beta} ... \int_{-\infty}^{x_{iT}\beta} \lambda(u_{i1}, u_{i2}...u_{iT}) du_{iT}...du_{i2}\ du_{i1} \end{aligned}$$

Solution?

$$\phi(u_{i1}, u_{i2}, ...u_{iT}) = \int_{-\infty}^{\infty} \phi(u_{i1}, u_{i2}, ...u_{iT} \mid \alpha_i)\phi(\alpha_i) d\alpha_i$$

Practical Things

Points:

- $\hat{\rho}$ = proportion of the variance due to the α_is.
- Implementation: Gauss-Hermite quadrature or MCMC.
- Best with N large and T small.
- Critically requires $\text{Cov}(\mathbf{X}, \alpha) = 0$ (see notes re: Chamberlain's CRE Estimator).

Example: WDI "Plus"

Data from the WDI, plus POLITY and the UCDP:

- `ISO3` - The country's International Standards Organization (ISO) three-letter identification code.
- `Year` - The year that row of data applies to (1960=1).
- `CivilWar` - Civil conflict indicator: 1 if there was a civil conflict in that country in that year; 0 otherwise. From the UCDP.
- `LandArea` - Land area (sq. km).
- `PopMillions` - Popluation (in millions).
- `PopGrowth` - Population Growth (percent).
- `UrbanPopulation` - Urban Population (percent of total).
- `GDPPerCapita` - GDP per capita (constant 2010 $US).
- `GDPPerCapGrowth` - GDP Per Capita Growth (percent annual).
- `PostColdWar` - 1 if `Year` $>$ 1989, 0 otherwise.
- `POLITY` - The POLITY score of democracy/autocracy. Scaled so that 0 = most autocratic, 10 = most democratic.

$N = 215$, $\bar{T} = 64$, NT varies (due to missingness).

Model and Data

Model:

$$\begin{aligned}\texttt{Civil War}_{it} \quad = \quad & f[\beta_0 + \beta_1 ln(\texttt{Land Area}_{it}) + \beta_2 ln(\texttt{Population}_{it}) + \\ & + \beta_3 \texttt{Urban Population}_{it} + \beta_4 ln(\texttt{GDP}_{it}) + \beta_5 \texttt{GDP Growth}_{it} + \\ & + \beta_6 \texttt{Post-Cold War}_{it} + \beta_7 \texttt{POLITY}_{it} + \beta_5 \texttt{POLITY}_{it}^2 + u_{it}]\end{aligned}$$

```
> describe(DF,skew=FALSE)
                vars     n      mean         sd    median     min        max       range       se
ISO3*              1 14252    108.47      62.35    108.00    1.00      216.0      215.00     0.52
Year*              2 14252     33.49      19.05     33.00    1.00       66.0       65.00     0.16
YearNumeric        3 14190   1992.50      19.05   1992.50 1960.00     2025.0       65.00     0.16
country*           4 14190    108.00      62.07    108.00    1.00      215.0      214.00     0.52
CivilWar           5  9378      0.13       0.34      0.00    0.00        1.0        1.00     0.00
LandArea           7 12288 596587.32 1645011.14 100540.00    2.03 16389950.0 16389947.97 14839.81
PopMillions        8 13945     25.54     106.13      4.33    0.00     1450.9     1450.93     0.90
UrbanPopulation    9 13975     52.21      25.80     52.12    1.84      100.0       98.16     0.22
GDPPerCapita      10 11307  12420.38   19440.07   3997.77  122.68   247170.2   247047.54   182.82
GDPPerCapGrowth   11 11184      1.91       6.54      2.10  -64.42      140.5      204.91     0.06
PostColdWar       12 14190      0.55       0.50      1.00    0.00        1.0        1.00     0.00
POLITY            13  8310      5.60       3.68      6.50    0.00       10.0       10.00     0.04
POLITYSquared     14  8310     44.95      40.04     42.25    0.00      100.0      100.00     0.44
```

Variation

Variable	Dim	Mean	SD	Min	Max	Observations
YearNumeric	overall	1992.5	19.051	1960	2025	N = 14190
	between		0	1992.5	1992.5	n = 215
	within		19.051	1960	2025	T = 66
CivilWar	overall	0.133	0.339	0	1	N = 9378
	between		0.222	0	1	n = 171
	within		0.253	-0.787	1.117	T = 54.842
LandArea	overall	596587.319	1645011.136	2.027	16389950	N = 12288
	between		1754315.837	2.03	16379186.875	n = 215
	within		5522.286	428120.652	651733.338	T = 57.153
PopMillions	overall	25.538	106.126	0.003	1450.936	N = 13945
	between		102.116	0.008	1113.738	n = 215
	within		29.352	-458.817	556.128	T = 64.86
UrbanPopulation	overall	52.209	25.796	1.839	100	N = 13975
	between		23.952	8.128	100	n = 215
	within		9.715	0.976	89.053	T = 65
GDPPerCapita	overall	12420.376	19440.071	122.679	247170.22	N = 11307
	between		21065.471	286.863	167187.157	n = 211
	within		7465.265	-40583.893	133748.603	T = 53.588
GDPPerCapGrowth	overall	1.908	6.541	-64.424	140.491	N = 11184
	between		1.692	-7.984	7.697	n = 212
	within		6.362	-66.273	134.702	T = 52.755
PostColdWar	overall	0.545	0.498	0	1	N = 14190
	between		0	0.545	0.545	n = 215
	within		0.498	0	1	T = 66
POLITY	overall	5.604	3.681	0	10	N = 8310
	between		2.967	0	10	n = 163
	within		2.219	-1.421	12.312	T = 50.982
POLITYSquared	overall	44.95	40.035	0	100	N = 8310
	between		33.014	0	100	n = 163
	within		22.853	-30.427	115.645	T = 50.982

Pooled Logit

```
> Logit<-glm(CivilWar~log(LandArea)+log(PopMillions)+UrbanPopulation+log(GDPPerCapita)+
+          GDPPerCapGrowth+PostColdWar+POLITY+POLITYSquared,data=DF,family="binomial")

> summary(Logit)

Call:
glm(formula = CivilWar ~ log(LandArea) + log(PopMillions) + UrbanPopulation +
    log(GDPPerCapita) + GDPPerCapGrowth + PostColdWar + POLITY +
    POLITYSquared, family = "binomial", data = DF)

Coefficients:
                  Estimate Std. Error z value Pr(>|z|)
(Intercept)       -0.20188    0.49219   -0.41  0.68169
log(LandArea)     -0.11586    0.03027   -3.83  0.00013 ***
log(PopMillions)   0.69996    0.03572   19.59  < 2e-16 ***
UrbanPopulation    0.01438    0.00294    4.90  9.6e-07 ***
log(GDPPerCapita) -0.44336    0.05406   -8.20  2.4e-16 ***
GDPPerCapGrowth   -0.03123    0.00602   -5.18  2.2e-07 ***
PostColdWar       -0.47639    0.08337   -5.71  1.1e-08 ***
POLITY             0.72826    0.06160   11.82  < 2e-16 ***
POLITYSquared     -0.06950    0.00578  -12.02  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 6010.2  on 7504  degrees of freedom
Residual deviance: 4929.1  on 7496  degrees of freedom
  (6747 observations deleted due to missingness)
AIC: 4947

Number of Fisher Scoring iterations: 6
```

Fixed Effects

```
> FELogit<-bife(CivilWar~log(LandArea)+log(PopMillions)+UrbanPopulation+log(GDPPerCapita)+
+              GDPPerCapGrowth+PostColdWar+POLITY+POLITYSquared|ISO3,data=DF,model="logit")

> summary(FELogit)
binomial - logit link

CivilWar ~ log(LandArea) + log(PopMillions) + UrbanPopulation +
    log(GDPPerCapita) + GDPPerCapGrowth + PostColdWar + POLITY +
    POLITYSquared | ISO3

Estimates:
                   Estimate Std. error z value Pr(> |z|)
log(LandArea)     -13.78997    9.31125   -1.48     0.139
log(PopMillions)    0.52131    0.26401    1.97     0.048 *
UrbanPopulation     0.01249    0.00903    1.38     0.167
log(GDPPerCapita)  -0.39393    0.15951   -2.47     0.014 *
GDPPerCapGrowth    -0.04406    0.00780   -5.65   1.6e-08 ***
PostColdWar        -0.11913    0.17288   -0.69     0.491
POLITY              0.70726    0.09291    7.61   2.7e-14 ***
POLITYSquared      -0.07424    0.00875   -8.49   < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

residual deviance= 2952,
null deviance= 4547,
n= 4214, N= 84

( 6747 observation(s) deleted due to missingness )
( 3291 observation(s) deleted due to perfect classification )

Number of Fisher Scoring Iterations: 6

Average individual fixed effect= 172.2
```

Alternative Fixed Effects (using `feglm`)

```
> FELogit2<-feglm(CivilWar~log(LandArea)+log(PopMillions)+UrbanPopulation+log(GDPPerCapita)+
+           GDPPerCapGrowth+PostColdWar+POLITY+POLITYSquared|ISO3,data=DF,family="binomial")
NOTES: 6,747 observations removed because of NA values (LHS: 4,874, RHS: 6,747).
       76 fixed-effects (3,291 observations) removed because of only 0 or only 1 outcomes
       or singletons.

> summary(FELogit2)
GLM estimation, family = binomial, Dep. Var.: CivilWar
Observations: 4,214
Fixed-effects: ISO3: 84
Standard-errors: IID
                   Estimate Std. Error z value   Pr(>|z|)
log(LandArea)     -13.78995   9.413465 -1.4649 1.4294e-01
log(PopMillions)    0.52132   0.266912  1.9531 5.0803e-02 .
UrbanPopulation     0.01249   0.009131  1.3678 1.7137e-01
log(GDPPerCapita)  -0.39393   0.161257 -2.4429 1.4571e-02 *
GDPPerCapGrowth    -0.04406   0.007888 -5.5864 2.3182e-08 ***
PostColdWar        -0.11913   0.174782 -0.6816 4.9548e-01
POLITY              0.70726   0.093930  7.5297 5.0855e-14 ***
POLITYSquared      -0.07424   0.008844 -8.3949  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Log-Likelihood: -1,475.8   Adj. Pseudo R2: 0.310836
           BIC:  3,719.4     Squared Cor.: 0.388961
```

Random Effects

```
> RELogit<-pglm(CivilWar~log(LandArea)+log(PopMillions)+UrbanPopulation+log(GDPPerCapita)+
+          GDPPerCapGrowth+PostColdWar+POLITY+POLITYSquared|ISO3,data=DF,family=binomial,
+          effect="individual",model="random",method="bfgs")

> summary(RELogit)
--------------------------------------------
Maximum Likelihood estimation
BFGS maximization, 206 iterations
Return code 0: successful convergence
Log-Likelihood: -1689
10  free parameters
Estimates:
                  Estimate Std. error t value      Pr(> t)
(Intercept)       -0.15113    1.04945   -0.14         0.89
log(LandArea)     -0.05012    0.05874   -0.85         0.39
log(PopMillions)   0.97490    0.07746   12.59      < 2e-16 ***
UrbanPopulation    0.00333    0.00508    0.66         0.51
log(GDPPerCapita) -0.67386    0.11253   -5.99 0.0000000021 ***
GDPPerCapGrowth   -0.04290    0.00755   -5.69 0.0000000130 ***
PostColdWar       -0.16196    0.12256   -1.32         0.19
POLITY             0.71733    0.08325    8.62      < 2e-16 ***
POLITYSquared     -0.07636    0.00808   -9.45      < 2e-16 ***
sigma              1.98996    0.09384   21.21      < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
--------------------------------------------
```

Nice Table...

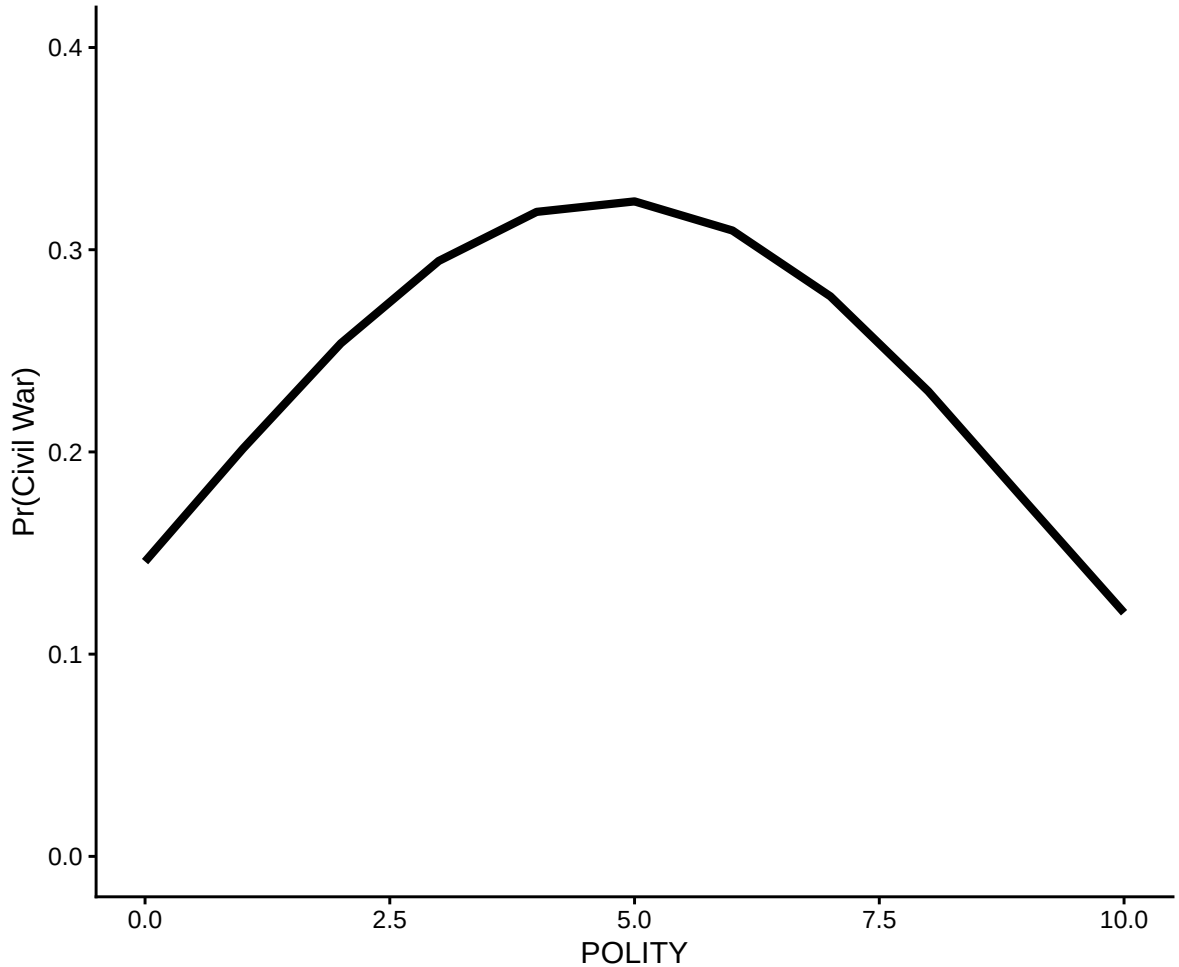
Models of Civil War

	Logit	FE Logit	FEs+Robust	RE Logit
Intercept	−0.17			−1.44
	(0.49)			(0.82)
ln(Land Area)	−0.12*	−13.77	−13.77	0.03
	(0.03)	(9.30)	(9.40)	(0.05)
ln(Population)	0.70*	0.52*	0.52	0.73*
	(0.04)	(0.26)	(0.27)	(0.06)
Urban Population	0.01*	0.01	0.01	0.02*
	(0.00)	(0.01)	(0.01)	(0.00)
ln(GDP Per Capita)	−0.45*	−0.39*	−0.39*	−0.65*
	(0.05)	(0.16)	(0.16)	(0.09)
GDP Growth	−0.03*	−0.04*	−0.04*	−0.04*
	(0.01)	(0.01)	(0.01)	(0.01)
Post-Cold War	−0.48*	−0.12	−0.12	−0.13
	(0.08)	(0.17)	(0.17)	(0.12)
POLITY	0.73*	0.71*	0.71*	0.70*
	(0.06)	(0.09)	(0.09)	(0.08)
POLITY Squared	−0.07*	−0.07*	−0.07*	−0.08*
	(0.01)	(0.01)	(0.01)	(0.01)
Estimated Sigma				2.11*
				(0.10)
AIC	4944.96			3394.35
BIC	5007.27			
Log Likelihood	−2463.48	−1475.46	−1475.46	−1687.18
Deviance	4926.96	2950.92	2950.92	
Num. obs.	7505	4214	4214	
Num. groups: ISO3			84	
Pseudo R^2			0.31	

* $p < 0.05$

Marginal Effects: POLITY

Predicted probabilities of `CivilWar`=1, by POLITY



Generalized Estimating Equations

Quick GLM review

Linear-normal model is:

$$Y_i = \mu_i + u_i$$

with:

$$\mu_i = \mathbf{X}_i \boldsymbol{\beta}.$$

Generalize:

$$g(\mu_i) = \mathbf{X}_i \boldsymbol{\beta}$$

and:

$$Y_i \sim \text{i.i.d. } F[\mu_i, \mathbf{V}_i].$$

"Score" equations:

$$\mathbf{U}(\beta) = \sum_{i=1}^{N} \mathbf{D}_i' \mathbf{V}_i^{-1} [Y_i - \mu_i] = \mathbf{0}.$$

with:

- $\mathbf{D}_i = \frac{\partial \mu_i}{\partial \beta}$,
- $\mathbf{V}_i = \frac{h(\mu_i)}{\phi}$, and
- $(Y_i - \mu_i) \approx$ a "residual."
- Known as "quasi-likelihood" (e.g. Wedderburn 1974 *Biometrika*).

GLMs $\rightarrow$ GEEs

Now suppose:

$$Y_{it} = \mu_{it} + u_{it}$$

where

- $i \in \{1, ...N\}$ are i.i.d. "units,"
- $t \in \{1, ...T\}$, $T > 1$ are "time points,"
- we want $g(\mu_{it}) = \mathbf{X}_{it}\boldsymbol{\beta}$.

Key issue: Accounting for (conditional) dependence in Y over time.

Full joint distributions over T are hard. But...

Define:

$$\underset{T\times T}{\mathbf{R}_i(\boldsymbol{\alpha})} = \begin{pmatrix} 1.0 & \alpha_{12} & \cdots & \alpha_{1,T} \\ \alpha_{21} & 1.0 & \cdots & \alpha_{2,T} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{T,1} & \cdots & \alpha_{T,T-1} & 1.0 \end{pmatrix},$$

$\rightarrow$ "working correlation" matrix.

- Completely defined by $\boldsymbol{\alpha}$,
- Structure specified by the analyst.

Liang and Zeger (1986): We can decompose the variance of Y_{it} as:

$$\mathbf{V}_i = \text{diag}(\mathbf{V}_i^{\frac{1}{2}})\,\mathbf{R}_i(\boldsymbol{\alpha})\,\text{diag}(\mathbf{V}_i^{\frac{1}{2}})$$

With a standard GLM assumption about the mean and variance, this is:

$$\mathbf{V}_i = \frac{(\mathbf{A}_i^{\frac{1}{2}})\,\mathbf{R}_i(\boldsymbol{\alpha})\,(\mathbf{A}_i^{\frac{1}{2}})}{\phi}$$

where

$$\underset{T\times T}{\mathbf{A}_i} = \begin{pmatrix} h(\mu_{i1}) & 0 & \cdots & 0 \\ 0 & h(\mu_{i2}) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & h(\mu_{iT}) \end{pmatrix}$$

What does that mean?

$\mathbf{V}_i = \text{Var}(Y_{it}|\mathbf{X}_{it}, \boldsymbol{\beta})$ has two parts:

- $\mathbf{A}_i =$ *unit-level* variation,
- $\mathbf{R}_i(\boldsymbol{\alpha}) =$ within-unit *temporal* variation.

Specifying $\mathbf{R}_i(\boldsymbol{\alpha})$

Independent: $$\mathbf{R}_i(\boldsymbol{\alpha}) = \begin{pmatrix} 1.0 & 0 & \cdots & 0 \\ 0 & 1.0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 1.0 \end{pmatrix}$$

- Assumes no within-unit temporal correlation.
- Equivalent to GLM on pooled data.

Exchangeable: $$\mathbf{R}_i(\boldsymbol{\alpha}) = \begin{pmatrix} 1.0 & \alpha & \cdots & \alpha \\ \alpha & 1.0 & \cdots & \alpha \\ \vdots & \vdots & \ddots & \vdots \\ \alpha & \cdots & \alpha & 1.0 \end{pmatrix}$$

- One free parameter in $\mathbf{R}_i(\boldsymbol{\alpha})$ ($\alpha_{ts} = \alpha \; \forall \; t \neq s$)
- Temporal correlation within units is constant across time points.
- Akin (in some respects) to a random-effects model...

Specifying $\mathbf{R}_i(\boldsymbol{\alpha})$

AR(p) (e.g., AR(1)): $\mathbf{R}_i(\boldsymbol{\alpha}) = \begin{pmatrix} 1.0 & \alpha & \alpha^2 & \cdots & \alpha^{T-1} \\ \alpha & 1.0 & \alpha & \cdots & \alpha^{T-2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \alpha^{T-1} & \cdots & \alpha^2 & \alpha & 1.0 \end{pmatrix}$

- One free parameter in $\mathbf{R}_i(\boldsymbol{\alpha})$ ($\alpha_{ts} = \alpha^{|t-s|} \; \forall \; t \neq s$).
- Conditional within-unit correlation an exponential function of the lag.

Stationary(p): $\mathbf{R}_i(\boldsymbol{\alpha}) = \begin{pmatrix} 1.0 & \alpha_1 & \cdots & \alpha_p & 0 & \cdots & 0 \\ \alpha_1 & 1.0 & \alpha_1 & \cdots & \cdots & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & \cdots & 0 & \alpha_p & \cdots & \alpha_1 & 1.0 \end{pmatrix}$

- AKA "banded," or "p-dependent."
- $p \leq T - 1$ free parameters in $\mathbf{R}_i(\boldsymbol{\alpha})$.
- Conditional within-unit correlation an exponential function of the lag, up to lag p, and zero thereafter.

Specifying $\mathbf{R}_i(\boldsymbol{\alpha})$

Unstructured: $\mathbf{R}_i(\boldsymbol{\alpha}) = \begin{pmatrix} 1.0 & \alpha_{12} & \alpha_{13} & \cdots & \alpha_{1,T-1} \\ \alpha_{12} & 1.0 & \alpha_{23} & \cdots & \alpha_{2,T-1} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \alpha_{1,T-1} & \alpha_{2,T-1} & \cdots & \alpha_{T-1,T-1} & 1.0 \end{pmatrix}$

- $\frac{T(T-1)}{2}$ free parameters in $\mathbf{R}_i(\boldsymbol{\alpha})$.
- Conditional within-unit correlation is completely data-dependent.

Score equations:

$$\boldsymbol{U}_{GEE}(\boldsymbol{\beta}_{GEE}) = \sum_{i=1}^{N} \mathbf{D}_i' \left[\frac{(\mathbf{A}_i^{\frac{1}{2}})\, \mathbf{R}_i(\boldsymbol{\alpha})\, (\mathbf{A}_i^{\frac{1}{2}})}{\phi} \right]^{-1} [Y_i - \mu_i] = \mathbf{0}$$

Two-step estimation:

- For fixed values of $\boldsymbol{\alpha}_s$ and ϕ_s at iteration s, use Newton scoring to estimate $\hat{\boldsymbol{\beta}}_s$,
- Use $\hat{\boldsymbol{\beta}}_s$ to calculate standardized residuals $(Y_i - \hat{\mu}_i)_s$, from which consistent estimates of $\boldsymbol{\alpha}_{s+1}$ and ϕ_{s+1} can be estimated.

Liang & Zeger (1986):

$$\hat{\boldsymbol{\beta}}_{GEE} \underset{N\to\infty}{\sim} \boldsymbol{N}(\boldsymbol{\beta}, \boldsymbol{\Sigma}).$$

For $\hat{\boldsymbol{\Sigma}}$, two options:

$$\hat{\boldsymbol{\Sigma}}_{\text{Model}} = N\left(\sum_{i=1}^{N} \hat{\boldsymbol{D}}_i' \hat{\boldsymbol{V}}_i^{-1} \hat{\boldsymbol{D}}_i\right)$$

$$\hat{\boldsymbol{\Sigma}}_{\text{Robust}} = N\left(\sum_{i=1}^{N} \hat{\boldsymbol{D}}_i' \hat{\boldsymbol{V}}_i^{-1} \hat{\boldsymbol{D}}_i\right)^{-1} \left(\sum_{i=1}^{N} \hat{\boldsymbol{D}}_i' \hat{\boldsymbol{V}}_i^{-1} \hat{\boldsymbol{S}}_i \hat{\boldsymbol{V}}_i^{-1} \hat{\boldsymbol{D}}_i\right) \left(\sum_{i=1}^{N} \hat{\boldsymbol{D}}_i' \hat{\boldsymbol{V}}_i^{-1} \hat{\boldsymbol{D}}_i\right)^{-1}$$

where $\hat{\boldsymbol{S}}_i = (Y_i - \hat{\mu}_i)(Y_i - \hat{\mu}_i)'$.

Inference (aka, magic!)

For GEEs:

- $\hat{\Sigma}_{\text{Model}}$
  - Requires that $\mathbf{R}_i(\boldsymbol{\alpha})$ be "correct" for consistency.
  - Is slightly more efficient than $\hat{\Sigma}_{\text{Robust}}$ if so.

- $\hat{\Sigma}_{\text{Robust}}$
  - Is consistent *even if* $\mathbf{R}_i(\boldsymbol{\alpha})$ *is misspecified*.
  - Is slightly less efficient than $\hat{\Sigma}_{\text{Model}}$ if $\mathbf{R}_i(\boldsymbol{\alpha})$ is correct.

Moral: Use $\hat{\Sigma}_{\text{Robust}}$.

Summary

GEEs:

- Are a straightforward variation on GLMs, and so
- Can be applied to a range of data types (continuous, binary, count, proportions, etc.),
- Yield robustly consistent point estimates of βs,
- Account for within-unit correlation in an informed way, but also
- Yield consistent inferences even if the correlation is misspecified.

Practical Issues: Model Interpretation

- In general, GEEs $=$ GLMs.
- GEEs are *marginal* models, so:
  - $\hat{\beta}$s have an interpretation as *average* / total effects.
  - Estimates / effect sizes generally be smaller than conditional (e.g. fixed/random) effects models.
  - E.g., for logit, $\hat{\boldsymbol{\beta}}_M \approx \frac{\hat{\boldsymbol{\beta}}_C}{\sqrt{1+0.35\sigma^2_\eta}}$, where $\sigma^2_\eta > 0$ is the variance of the unit effects.
  - See (e.g.) Gardiner et al. (2009) or Koper and Manseau (2009) for expositions.

Practical Issues: Specifying $\mathbf{R}_i(\boldsymbol{\alpha})$

- Has been called "more art than science."
- Pointers:
  - Choose based on *substance* of the problem.
  - Remember that $\mathbf{R}_i(\boldsymbol{\alpha})$ is conditional on $\mathbf{X}$, $\hat{\boldsymbol{\beta}}$.
  - Consider unstructured when T is small and N large.
  - Try different ones, and compare.
- In general, it shouldn't matter terribly much...

GEEs: Software Tips

Tips:

- Generally follow GLMs (specify "family" + "link")
- Certain combinations not possible/recommended
- Estimation: Fisher scoring, MLE, etc. (MCMC?)

From the `geepack` manual:

Warning

Use "unstructured" correlation structure only with great care. (It may cause R to crash).

GEE: Independence (2013-2017 only)

```
> GEE.ind5<-geeglm(CivilWar~log(LandArea)+log(PopMillions)+UrbanPopulation+
+                    log(GDPPerCapita)+GDPPerCapGrowth+POLITY+
+                    POLITYSquared,data=DF5,id=ISO3,family="binomial",
+                  corstr="independence")
> summary(GEE.ind5)

Call:
geeglm(formula = CivilWar ~ log(LandArea) + log(PopMillions) +
    UrbanPopulation + log(GDPPerCapita) + GDPPerCapGrowth + POLITY +
    POLITYSquared, family = "binomial", data = DF5, id = ISO3,
    corstr = "independence")

 Coefficients:
                 Estimate Std.err  Wald Pr(>|W|)
(Intercept)       -3.6488  4.0195  0.82  0.36400
log(LandArea)      0.0206  0.1925  0.01  0.91473
log(PopMillions)   0.8628  0.2373 13.22  0.00028 ***
UrbanPopulation    0.0252  0.0179  1.99  0.15828
log(GDPPerCapita) -0.4773  0.3409  1.96  0.16149
GDPPerCapGrowth   -0.0203  0.0345  0.35  0.55582
POLITY             0.8835  0.5505  2.58  0.10849
POLITYSquared     -0.0901  0.0472  3.65  0.05611 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Correlation structure = independence
Estimated Scale Parameters:

            Estimate Std.err
(Intercept)    0.631   0.279
Number of clusters:   157  Maximum cluster size: 5
```

GEE: Exchangeable (2013-2017)

```
> GEE.exc5<-geeglm(CivilWar~log(LandArea)+log(PopMillions)+UrbanPopulation+
+                    log(GDPPerCapita)+GDPPerCapGrowth+POLITY+
+                    POLITYSquared,data=DF5,id=ISO3,family="binomial",
+                  corstr="exchangeable")
> summary(GEE.exc5)

Call:
geeglm(formula = CivilWar ~ log(LandArea) + log(PopMillions) +
    UrbanPopulation + log(GDPPerCapita) + GDPPerCapGrowth + POLITY +
    POLITYSquared, family = "binomial", data = DF5, id = ISO3,
    corstr = "exchangeable")

 Coefficients:
                  Estimate  Std.err  Wald Pr(>|W|)
(Intercept)       -2.38008  3.45343  0.47  0.49070
log(LandArea)      0.03679  0.20356  0.03  0.85659
log(PopMillions)   0.81950  0.23636 12.02  0.00053 ***
UrbanPopulation    0.02714  0.01689  2.58  0.10808
log(GDPPerCapita) -0.58020  0.29085  3.98  0.04606 *
GDPPerCapGrowth   -0.00784  0.02228  0.12  0.72492
POLITY             0.50190  0.42977  1.36  0.24287
POLITYSquared     -0.05319  0.03739  2.02  0.15491
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Correlation structure = exchangeable
Estimated Scale Parameters:

            Estimate Std.err
(Intercept)    0.635   0.326
  Link = identity

Estimated Correlation Parameters:
      Estimate Std.err
alpha    0.503    0.27
Number of clusters:   157  Maximum cluster size: 5
```

GEE: Unstructured (2013-2017)

```
> GEE.unstr<-geeglm(CivilWar~log(LandArea)+log(PopMillions)+UrbanPopulation+log(GDPPerCapita)+GDPPerCapGrowth+
+                     POLITY+POLITYSquared,data=DF5,id=ISO3,family="binomial",corstr="unstructured")
> summary(GEE.unstr)

Call:
geeglm(formula = CivilWar ~ log(LandArea) + log(PopMillions) +
    UrbanPopulation + log(GDPPerCapita) + GDPPerCapGrowth + POLITY +
    POLITYSquared, family = "binomial", data = DF5, id = ISO3,
    corstr = "unstructured")

 Coefficients:
                  Estimate Std.err  Wald Pr(>|W|)
(Intercept)        -2.7362  3.1656  0.75  0.38739
log(LandArea)       0.0182  0.1888  0.01  0.92311
log(PopMillions)    0.8525  0.2365 12.99  0.00031 ***
UrbanPopulation     0.0308  0.0155  3.96  0.04663 *
log(GDPPerCapita)  -0.5698  0.2882  3.91  0.04806 *
GDPPerCapGrowth     0.0265  0.0274  0.93  0.33505
POLITY              0.5791  0.4242  1.86  0.17217
POLITYSquared      -0.0608  0.0377  2.60  0.10670
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Correlation structure = unstructured
Estimated Scale Parameters:

            Estimate Std.err
(Intercept)    0.663   0.376
  Link = identity

Estimated Correlation Parameters:
          Estimate Std.err
alpha.1:2    0.569   0.310
alpha.1:3    0.531   0.306
alpha.1:4    0.439   0.276
alpha.1:5    0.277   0.187
alpha.2:3    0.856   0.439
alpha.2:4    0.380   0.258
alpha.2:5    0.383   0.247
alpha.3:4    0.485   0.319
alpha.3:5    0.607   0.367
alpha.4:5    0.439   0.293
Number of clusters:   157  Maximum cluster size: 5
```

GEE Model Comparison (2013-2017)

Table 4: GEE Models of Civil War Onset

	Independence	Exchangeable	AR(1)	Unstructured
(Intercept)	−3.649	−2.380	−4.399	−2.736
	(4.020)	(3.453)	(3.736)	(3.166)
ln(Land Area)	0.021	0.037	0.148	0.018
	(0.192)	(0.204)	(0.208)	(0.189)
ln(Population)	0.863***	0.820***	0.732**	0.852***
	(0.237)	(0.236)	(0.251)	(0.237)
Urban Population	0.025	0.027	0.021	0.031*
	(0.018)	(0.017)	(0.017)	(0.016)
ln(GDP Per Capita)	−0.477	−0.580*	−0.445	−0.570*
	(0.341)	(0.291)	(0.300)	(0.288)
GDP Growth	−0.020	−0.008	0.026	0.026
	(0.035)	(0.022)	(0.036)	(0.027)
POLITY	0.884	0.502	0.524	0.579
	(0.550)	(0.430)	(0.517)	(0.424)
POLITY Squared	−0.090+	−0.053	−0.056	−0.061
	(0.047)	(0.037)	(0.047)	(0.038)
Num.Obs.	781	781	781	781

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

GEEs: Wrap-Up

GEEs are:

- Robust
- Flexible
- Extensible beyond panel/TSCS context

GEEs: References

A few useful resources:

- The GEE Wikipedia page
- Liang and Zeger (1986)
- Vanegas et al. (2023): "Generalized Estimating Equations Using the New R Package `glmtoolbox`."
- Textbooks, etc.:
  - Diggle et al. (2002)
  - Zeigler (2011)
  - Hardin and Hilbe (2013)

Software

R:

- `pglm` (panel GLMs) (maximum likelihood + quadrature)
- `bife` (fixed-effects logit / probit only)
- `glmer` (general mixed-effects models, including RE)
- GEEs: `geepack`, `gee`, `multgee`

Python:

- `PyFixEst` (fixed effects models)
- `statsmodels` (conditional models / GEE)
- `PyMC` (Bayesian hierarchical models)

Stata:

- `xtprobit`, `xtlogit`, `xtcloglog`
- Plus `xttrans` (transition probabilities), `quadchk` (quadrature checking), `xtrho` / `xtrhoi` (estimation of within-unit covariances)
- `xtgee` (or `xtreg...,pa`) for GEEs